Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce Department of Information Technology

(B.Sc.I.T Semester V)

Full Stack Development using MERN

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1. Project Title

StudySphere – Student Study Planner and Progress Tracker Using

2. Objective

The main objective of StudySphere is to develop a full-stack web application using React, Node.js, Express.js, and MongoDB.The application helps students organize their academic tasks, manage assignment deadlines, and track their study progress.The project also demonstrates practical knowledge of CRUD operations, REST API integration, database connectivity, form validation, and error handling.

3. Project Description

StudySphere is a web-based student study planner designed to help students manage their daily academic activities in an organized way.

The application allows students to add study tasks with details such as task title, subject, description, priority, due date, and status. Students can view, edit, delete, complete, and search their tasks.

It also provides a dashboard and progress tracker to display completed, pending, and ongoing tasks. The frontend is developed using React, while Node.js and Express.js handle backend operations. MongoDB is used to store task information, and Axios connects the frontend with the backend APIs.

4. Functional Requirements

Feature	Requirements
Add Records	Students can add new study tasks through a form and save them in MongoDB.
View Records	Saved tasks are retrieved from the database and displayed in a table.
Update Records	Students can edit task details such as title, subject, priority, status, and due date.
Delete Records	Students can delete selected tasks after confirmation.
Search	Students can search tasks using relevant fields such as title, subject, priority, or status.
Form Validation	Required fields and data formats are validated before saving records.
Error Handling	Suitable success and error messages are displayed during operations.
Progress Tracking	The application displays task completion progress and status information.

5. Tools and Technologies Used

- React
- Node.js
- Express.js
- MongoDB
- Mongoose
- Axios
- Visual Studio Code ● Git and GitHub

6. Technical Requirements

Component	Requirement
Frontend	React.js is used to create the dashboard, task form, and task table.
React Hooks	useState() manages application state, and useEffect() loads task data.

API Integration	Axios sends HTTP requests between frontend and backend.
Backend	Node.js and Express.js are used to create REST APIs.
Database	MongoDB and Mongoose store and manage task records.
CRUD Operations	GET, POST, PUT, and DELETE methods are used for task management.
Validation	Input validation is implemented on both frontend and backend.

7. Validation Requirements

- **Required Field Validation:** Task title, subject, and due date must not be empty.
- **Text Validation:** Task title and subject must follow the defined length requirements.
- **Date Validation:** Due dates must follow a valid date format.
- **Priority and Status Validation:** Only permitted priority and status values are accepted.
- **Duplicate Handling:** Different assignments may have the same title, so duplicate task titles are allowed.
- **Error Handling:** Appropriate messages are displayed for invalid input or API errors.
- **Delete Confirmation:** The application asks for confirmation before deleting a task.

8. Testing and Demonstration

The StudySphere application is tested to check its major functionalities, including adding, viewing, updating, deleting, and searching study tasks. Form validation is checked by entering empty or invalid details, while error handling is checked for unsuccessful operations.

During the demonstration, the user can add a new task, view saved tasks, edit task details, mark tasks as completed, search for specific tasks, and delete tasks after confirmation. The dashboard and progress tracker display the status of academic tasks.

9. GitHub Repository and Submission

Students must upload their complete project source code to GitHub and submit the repository link.

GitHub Repository Link: https://github.com/vinapadhmvlu-hue/mern-project